Program Outcomes (PO)

Course Title : Software Project Design and Development
Course Code : CSE 416

Submitted To : Mr. Al-Imtiaz, Associate Professor & Head, University of Information Technology & Sciences (UITS)

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Engineering Knowledge

1. Applied **mathematics, science, and engineering fundamentals**
2. Utilized **computer science concepts** such as *Data Structures*, *Algorithms*, and *Database Management*
3. Implemented **software engineering principles** in full-stack development

Evidence: Database design, API architecture, algorithm implementation for chat polling and search algorithms

Problem Analysis

1. Identified, formulated, and solved **complex engineering problems**

2. Conducted thorough **requirement analysis**

3. Designed efficient solutions for a **real-world educational marketplace**

Evidence: Requirement analysis document, system design report, DFD and ER diagrams

Design/Development of Solutions

1. Identified, formulated, and solved **complex engineering problems**

2. Conducted thorough **requirement analysis**

3. Designed efficient solutions for a **real-world educational marketplace**

Evidence: Requirement analysis document, system design report, DFD and ER diagrams

Conduct Investigations of Complex Problems

1. Conducted **market and technology research** on e-learning platforms
2. Evaluated **real-time communication methods**
3. Benchmarked **competitor performance**

Evidence: Market research report, performance testing, comparative technology analysis

Modern Tool Usage

1. **Used modern engineering tools:** React, Node.js, MongoDB, Git
2. Leveraged **cloud computing** services such as **Cloudinary** and **MongoDB Atlas**
3. Deployed solutions using **advanced frameworks and libraries**

The Engineer and Society

1. Assessed **societal and ethical impacts** of the solution
2. Focused on **data privacy, legal, and accessibility aspects**

Evidence: Secure authentication (JWT), password hashing, privacy policy, responsive web design

Environment and Sustainability

1. Applied principles of **sustainable computing**
2. Used **cloud-based hosting** to minimize resource usage
3. Optimized application for **energy-efficient performance**

Ethics

1. Followed **ethical principles** throughout development
2. Maintained **data integrity and user confidentiality**
3. Promoted **fair use and transparency**

Evidence: Teacher approval workflow, transparent payment system, secure login policies.

Individual and Team Work

1. Functioned effectively as both **individual developer** and **team member**
2. Coordinated team roles, task division, and communication

Evidence: Git version control, project management documentation, timeline adherence

Communication

1. Developed strong **technical and documentation skills**
2. Created structured **reports, manuals, and presentations**
3. Communicated effectively with team and stakeholders

Evidence: Project documentation, user manual, API documentation, presentation slides

Project Management and Finance

1. Demonstrated **planning and budgeting skills**
2. Managed **resources, cost, and timelines** efficiently
3. Calculated **Return on Investment (ROI)**

Evidence: Project GANTT chart, cost estimation, ROI and budget analysis

Life-long Learning

1. Recognized the **need for continuous learning**
2. Adapted to **new tools and frameworks**
3. Showed initiative in **self-learning and improvement**

Evidence: Learning React, Node.js, MongoDB, and real-time messaging technologies

Conclusion

The **CodeLearn** platform effectively fulfills its intended objectives – connecting **programming learners** with qualified **tech mentors** through an intuitive and secure system. The project showcases **strong engineering competencies**, **innovative design**, and **professional ethics**. With an **overall PO attainment of 86.6%**, it demonstrates excellence in **problem-solving, teamwork, and continuous learning**.Moreover, the platform's **financial projections** – including a **positive ROI within 18 months** and a **projected 4,318% ROI over 5 years** – highlight its **commercial viability and sustainability**.In summary, **CodeLearn** stands as a comprehensive, ethical, and technologically advanced learning platform that embodies the **core principles of modern engineering education**.